

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Calculus II

Semester: Spring

Year : 2024
Full Marks: 100
Pass Marks: 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Evaluate the integral by reversing the order of integration: 5
$$\int_0^{\pi} \int_x^{\pi} \left(\frac{\sin y}{y} \right) dy dx$$
 - b) Evaluate the integral: $\int_0^1 \int_0^{1-x} \int_0^{x+y} e^z dz dy dx$ 5
 - c) Find the volume of the solid whose base is the region in the xy -plane that is bounded by the parabola $y = 4 - x^2$ and the line $y = 3x$, while the top of the solid is bounded by the plane $z = x + 4$. 5
 2. a) Solve by using power series method: $y'' - 4xy' + (4x^2 - 2)y = 0$. 7
 - b) Solve the Bessel's equation $x^2 y'' + xy' + (x^2 - n^2)y = 0$. 8
- OR**
- i. Express $x^2 - 4x + 2$ as a Legendre polynomial by using Rodrigues formula.
 - ii. Sketch the graph of $P_0(x)$, $P_1(x)$, $P_2(x)$ and $P_3(x)$ in a single graph.
3. a) Solve the initial value problem by using Laplace transform: 7
$$y'' + 2y' + 2y = 5 \sin t, y(0) = y'(0) = 0$$
 - b) i. Find the Laplace transform of: $\cosh at \sin at$. 8
 - ii. Find the inverse Laplace transform of $\frac{s}{(s^2+1)(s^2+4)}$ using the convolution theorem.
4. a) If $\phi = x^3 + y^3 + z^3 - 3xyz$, find: $\text{div}(\text{grad} \phi)$ 5
 - b) Find the directional derivative of $\phi = x^2 - y^2 + 2z^2$ at the point $P(1, 2, 3)$ in the direction of line PQ, where Q is the point $(5, 0, 4)$. 5
 - c) Find the work done by the force $\vec{F} = (2y+3)\vec{i} + xz\vec{j} + (yz-x)\vec{k}$ when a particle moves from $(0, 0, 0)$ to the point $(2, 1, 1)$ along the curve $x = 2t^2, y = t, z = t^3$. 5

5. a) Find the flux of $\vec{F} = z\vec{i} + x\vec{j} + y\vec{k}$ over the portion of the surface $x^2 + y^2 = 1$ in the first octant between $z = 0$ to $z = 2$. 7
- b) State Stoke's theorem. Use it to evaluate $\oint_c \vec{F} \cdot d\vec{r}$, where $\vec{F} = (z, x, y)$ and c is the surface of upper half of sphere $x^2 + y^2 + z^2 = a^2$ in xy -plane. 8

OR

State Gauss divergence theorem. Evaluate $\iint_S \vec{F} \cdot \vec{n} \, ds$, where $\vec{F} = (4x, x^2y, -x^2z)$ and S is the surface of the tetrahedron with vertices $(0,0,0)$, $(1,0,0)$, $(0,1,0)$, and $(0,0,1)$, by using the Gauss divergence theorem.

6. a) Find the Fourier series of the function $f(x) = |x|$ for $-2 < x < 2$. 7
- b) Expand the function $f(x) = x^2$ in the interval $0 < x < \pi$ in half-range Fourier cosine series and deduce that $\frac{\pi^2}{6} = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots$. 8

7. Solve: (Any two) 2×5
- a) Find the general solution of $u_x + u_y - u = 0$.
- b) Derive one dimensional traffic flow model using convection law.
- c) Show that the integral $\int_{(0,0,0)}^{(2,4,0)} e^{x-y+z^2} (dx - dy + 2zdz)$ is exact and evaluate the exact value.